RUGVED VYAS

AI/ML Engineer | Generative AI, Agentic Systems & Full-Stack Development

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PROFESSIONAL SUMMARY

AI/ML Engineer and M.Tech candidate (CGPA 9.2) specializing in Generative AI, Computer Vision, and full-stack development. Skilled in building and deploying production-grade ML systems, from real-time YOLOv8 vision pipelines to LLM-powered web applications, using PyTorch, TensorFlow, Docker, and AWS. Engineered an AI-driven, xAPI-compliant course creation tool that integrates across all major LMS platforms. National-level hackathon finalist with a proven record of shipping scalable, end-to-end AI solutions and mentoring 100+ students.

CORE COMPETENCIES

- Generative AI and Prompt Engineering
- Computer Vision (YOLO, ResNet, CNNs)
- Predictive Analytics and Forecasting
- Multi-Agent Systems and Agentic AI
- Deep Learning and Neural Networks
- MLOps and Model Deployment
- Full-Stack Development and REST APIs
- Cloud Deployment and Edge Computing

TECHNICAL SKILLS

AI & ML FRAMEWORKS

PyTorch | TensorFlow | Scikit-Learn | Hugging Face | YOLOv8 | OpenCV | Keras

GENERATIVE AI & LLMS

OpenAI GPT | Gemini | Prompt Engineering | LLM Integration

FULL-STACK DEVELOPMENT

React | JavaScript | HTML | CSS | Node.js | Flask | FastAPI | REST APIs

CLOUD & DEVOPS

AWS S3 | AWS EC2 | Docker | MLflow | Git | CI/CD

LANGUAGES & DATABASES

Python (Expert) | C | C++ | SQL | MySQL | PostgreSQL | MongoDB

DATA SCIENCE & VIZ

Pandas | NumPy | Matplotlib | Seaborn | Plotly | EDA

ML ALGORITHMS

Random Forest | XGBoost | CNNs | RNNs | Transformers | Attention | SVM | SVR | Ensembles | KNN | K-Means

COMPUTER VISION & EDGE

Object Detection | Image Classification | Facial Recognition | Raspberry Pi | Edge ML | IoT

PROFESSIONAL EXPERIENCE

AI Full-Stack Developer Intern

Jan 2026 - Present

emPower eLearning, Ahmedabad, Gujarat

- Engineered an **AI-powered, xAPI-compliant course creation tool** that auto-generates structured e-learning content and integrates seamlessly across all major LMS platforms with full learner-progress tracking.
- Build and deploy full-stack LMS features, integrating ML models with React frontends and Python (Flask, FastAPI) backends via RESTful APIs.
- Develop AI-driven learning tools such as automated quiz generation, content summarization, and personalized recommendations, using GPT and Gemini.
- Containerize services with Docker and architect AWS S3 storage workflows; collaborate in an agile team following CI/CD and Git practices.

Teaching Assistant, Machine Learning & AI

Aug 2024 - Jan 2026

Ganpat University, Mehsana, Gujarat

- Instructed 100+ B.Tech students in Data Structures, Algorithms, and Machine Learning, designing hands-on labs on supervised and unsupervised learning, neural networks, and model evaluation.
- Mentored students across classification, regression, clustering, and computer vision projects, simplifying concepts like backpropagation, gradient descent, CNNs, and transformers.

Vice-President, Computer Society & Gaming Club

2022 – 2024

Institute of Advanced Research, Gandhinagar

- Led a 100+ member technical club, organizing hackathons, coding competitions, and workshops on Generative AI, LLMs, and Computer Vision.
- Built a knowledge-sharing culture that increased technical engagement by 40%.

KEY AI/ML PROJECTS & RESEARCH

Occlusion-Aware Citrus Fruit Detection & Ripeness Classification

Computer Vision, Deep Learning, Precision Agriculture

- Architected an end-to-end YOLOv8 system for real-time inference (30+ FPS), tackling fruit occlusion, dense foliage, and illumination variance.
- Integrated a ResNet-50 CNN for multi-class ripeness classification with occlusion-aware augmentations; evaluated using mAP and F1-score.

PyTorch | YOLOv8 | ResNet-50 | OpenCV | Albumentations

Rose Leaf Disease Detection Using Deep Learning

Computer Vision, Image Classification, Transfer Learning

- Built a deep learning system to identify rose leaf diseases (Black Spot, Downy Mildew, Healthy) for real-world field conditions.
- Benchmarked ResNet50, MobileNetV2, and EfficientNetB0; EfficientNetB0 achieved about 97% accuracy and 97% F1-score.

TensorFlow | Keras | EfficientNet | ResNet50 | MobileNetV2

AdVision: Advertising Campaign Success Prediction

Predictive Analytics, Business Intelligence, MLOps

- Built a predictive pipeline on 10K+ records with rich feature engineering; benchmarked 8+ algorithms, with Random Forest beating baselines by 15%.
- Delivered actionable insights on high-ROI channels, budget allocation, and audience segmentation.

Python | Scikit-Learn | XGBoost | Pandas | Seaborn

SmartSurv: Low-Cost Multi-Camera Real-Time Surveillance

Edge Computing, IoT, Real-Time Systems, **National Finalist**

- Designed a cost-efficient Raspberry Pi 4 system monitoring 4+ camera feeds with under 20ms latency.
- Engineered frame-differencing and background-subtraction motion detection at 90%+ accuracy. **Top Team in Gujarat, Code Unnati Marathon 2023.**

Python | OpenCV | Raspberry Pi | Multi-threading | Gmail API

Smart Career Recommendation System

Recommendation Engine, NLP, Feature Engineering, **Hackathon Runner-up**

- Developed a hybrid (content-based plus collaborative) recommendation engine at 72% accuracy with multi-modal TF-IDF feature extraction.
- Runner-up at SSIP Hackathon 2022 among 50+ teams for a practical EdTech application.

Python | Scikit-Learn | Flask | TF-IDF | Cosine Similarity

EDUCATION

M.Tech in Computer Engineering

2025 – Present, CGPA 9.2

U V Patel College of Engineering, Ganpat University

Specialization: Advanced AI/ML, Deep Learning, NLP, Computer Vision. Research: Generative AI and Multi-Agent Architectures.

B.Tech in Computer Science & Engineering

2020 – 2024, CGPA 7.51

Institute of Advanced Research, Gandhinagar

Coursework: Machine Learning, Data Structures, Algorithms, Database Systems. Capstone: Real-Time Multi-Camera Surveillance System.

ADDITIONAL EXPERIENCE & ACHIEVEMENTS

- **Finalist, Code Unnati Innovation Marathon 2023:** Top Team in Gujarat (National Level); built SmartSurv in a 72-hour sprint.
- **Runner-up, SSIP Hackathon 2022:** ML-based Smart Career Recommendation System among 50+ teams.
- **Teaching and Mentorship:** Guided 100+ students and delivered 20+ technical workshops on modern AI.
- **Exploratory Projects:** BERT Sentiment Analysis (85%), CNN on CIFAR-10 (89%), LSTM stock forecasting, and anomaly detection with K-Means, DBSCAN, and Isolation Forest.